Society of Engineering Science 51st Annual Technical Meeting

1–3 October 2014

Purdue University, West Lafayette, Indiana, USA

Equilibrium and kinetics of the DNA overstretching transition

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ABSTRACT

It is well known that the dsDNA molecule undergoes a phase transition from B-DNA into an overstretched state at high forces. For long, the structure of the overstretched state remained unknown and highly debated, but recent advances in experimental techniques have presented evidence of more than one possible phase (or even a mixed phase) depending on ionic conditions, temperature and base-pair sequence. Here, we present a theoretical model to study the overstretching transition with the possibility that the overstretched state is mixture of two phases: a structure with portions of inner strand separation (melted DNA (M-DNA)) and an extended phase that retains the base-pair structure (S-DNA). We model the dsDNA as a chain composed of n segments of length l, where the transition is studied by means of a Landau quartic potential with statistical fluctuations. The length l is a measure of cooperatives of the transition and is a key to characterize the overstretched phase. By analyzing the different values of l corresponding to a wide spectrum of experiments, we find that for a range of temperatures and ionic conditions, the overstretched form is likely to be a mix of M-DNA and S-DNA. For a transition close to a pure S-DNA state, where the change in extension is close to 1.7´ the original B-DNA length, we find l is ~25 base-pairs regardless of temperature and ionic concentration. Our model is fully analytical, yet it accurately reproduces the force-extension curves, as well as the transient kinetic behavior, seen in DNA overstretching experiments.